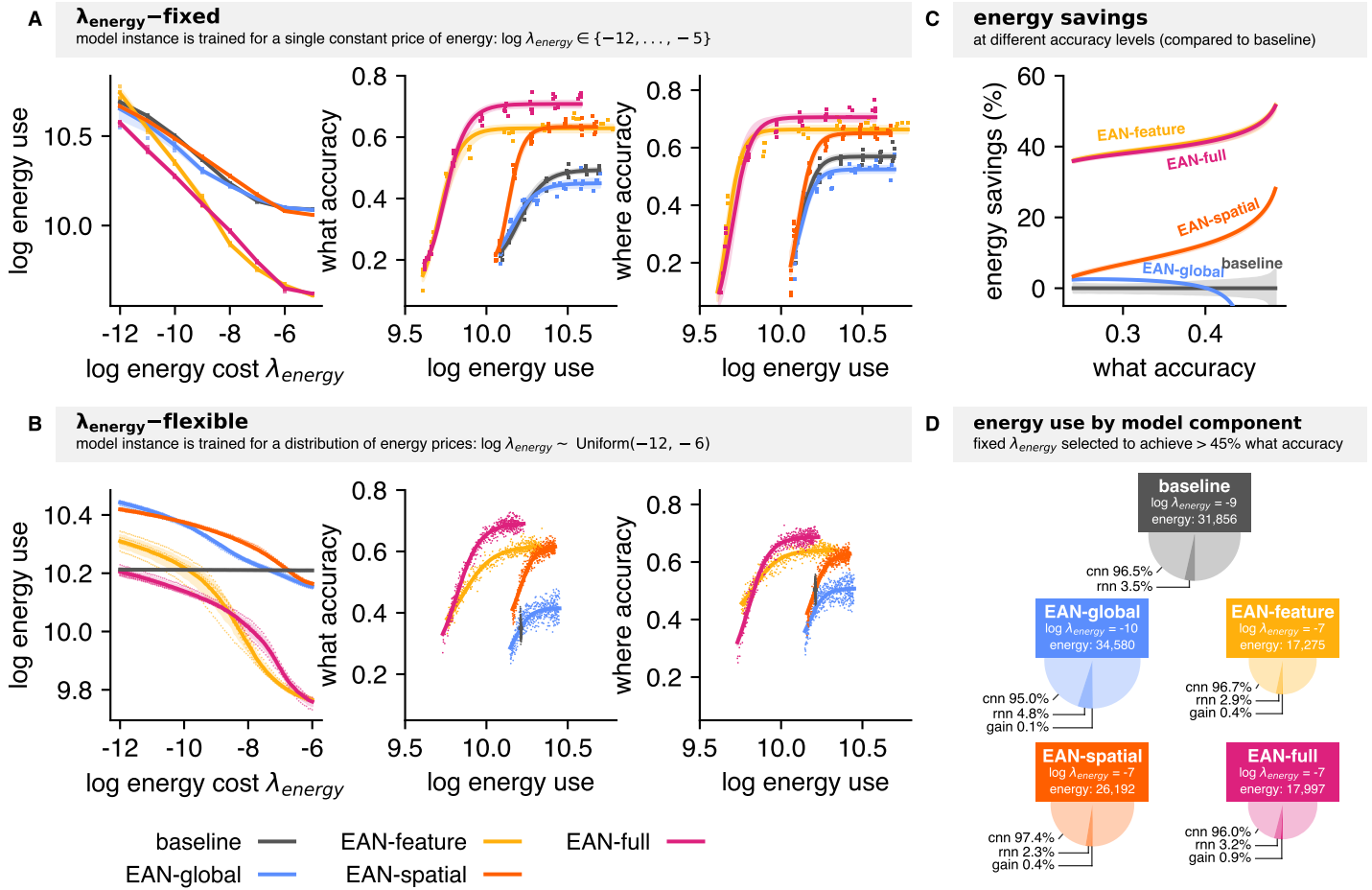
A
λenergy−fixed
model instance is trained for a single constant price of energy: log λenergy ∈ {−12, . . . , −5}
log energy use
10.5
10.0
-12
-10
-8
-6
log energy cost λenergy
what accuracy
0.8
0.6
0.4
0.2
9.5
10.0
10.5
log energy use
where accuracy
0.8
0.6
0.4
0.2
9.5
10.0
10.5
log energy use
B
λenergy−flexible
model instance is trained for a distribution of energy prices: log λenergy ∼ Uniform(−12, −6)
log energy use
10.4
10.2
10.0
9.8
-12
-10
-8
-6
log energy cost λenergy
what accuracy
0.8
0.6
0.4
0.2
9.5
10.0
10.5
log energy use
where accuracy
0.8
0.6
0.4
0.2
9.5
10.0
10.5
log energy use
baseline
EAN-feature
EAN-full
EAN-global
EAN-spatial
C
energy savings
at different accuracy levels (compared to baseline)
energy savings (%)
60
40
20
0
EAN-feature
EAN-full
EAN-spatial
EAN-global
baseline
0.3
0.4
what accuracy
D
energy use by model component
fixed λenergy selected to achieve > 45% what accuracy
baseline
log λenergy = -9
energy: 31,856
cnn 96.5%
rnn 3.5%
EAN-global
log λenergy = -10
energy: 34,580
cnn 95.0%
rnn 4.8%
gain 0.1%
EAN-feature
log λenergy = -7
energy: 17,275
cnn 96.7%
rnn 2.9%
gain 0.4%
EAN-spatial
log λenergy = -7
energy: 26,192
cnn 97.4%
rnn 2.3%
gain 0.4%
EAN-full
log λenergy = -7
energy: 17,997
cnn 96.0%
rnn 3.2%
gain 0.9%